

# ADITYA BAYU PRATAMA

## Supervisor MES & IT Engineer

SMART MANUFACTURING · INDUSTRY 4.0 · ENTERPRISE SYSTEMS

7+

YEARS IN SMART  
MANUFACTURING

500+

PRODUCT MODELS  
GOVERNED BY MES

5

ENGINEERS LED  
& MENTORED

2×

GLOBAL 1ST PLACE  
VE AWARD

### CONTACT

#### LOCATION

Batam, Riau Islands, Indonesia

#### PHONE

+62 882 7145 2089

#### EMAIL

[adityabayupratama13@gmail.com](mailto:adityabayupratama13@gmail.com)

#### PORTFOLIO

[adityabayupratama13.github.io/CV](https://adityabayupratama13.github.io/CV)

#### LINKEDIN

[linkedin.com/in/aditya-bayu-pratama-00561314a](https://linkedin.com/in/aditya-bayu-pratama-00561314a)

### CORE COMPETENCIES

#### SOFTWARE

C# (.NET), SQL / MySQL, Python, React.js, JavaScript, C++, LabVIEW, Dart / Flutter, REST APIs

#### SMART MANUFACTURING

MES architecture, Industrial IoT, traceability, Pokayoke, AOI machine vision, SMT process, Node-RED

#### HARDWARE

PLC (BNSP certified), SCADA, Raspberry Pi, Arduino, Lidar, data acquisition, PID control

#### AI & TOOLING

AI-assisted development, prompt engineering, LLM integration, LM Studio, Claude Code, Cursor, Codex, OpenCV

#### QUALITY

FMEA, DMAIC, Six Sigma practices, 5M change management, value engineering, BOM & QCPC, NPI

#### LEADERSHIP

Team recruitment, department budget, network infrastructure, server maintenance, incident response

### PROFILE

Supervisor MES & IT Engineer with more than seven years designing and running Manufacturing Execution Systems, Industrial IoT and machine vision for high-precision electronics and oil & gas. Grew the software and IT function from a solo role into a five-engineer department, architected a custom C#/.NET MES governing 500+ product models, and turned manual factory workflows into digital ones — one of them from 5–7 hours with 4–6 people to a 30-minute automated flow.

### EXPERIENCE

#### GIKEN PRECISION INDONESIA

Batam · Apr 2019 – Present

##### Supervisor MES & IT Engineer

Jul 2026 – Present

- Lead the MES and IT engineering function: system roadmap, architecture standards, project prioritisation across production and support, department budget, and development of the engineering team.
- Own the Giken One platform strategy — MES, traceability, feeder asset control and the internal web applications the plant depends on.
- Report department performance to management: uptime, ticket resolution, project progress and savings from digitalisation.

##### Lead Software & IT Engineer

Apr 2019 – Jun 2026

- Designed and deployed a custom Windows-based Manufacturing Execution System (C#/.NET, MySQL) governing 500+ product models across automotive, medical and consumer electronics lines, enforcing Pokayoke defect prevention and CTQ quality compliance.
- Built Giken One, an enterprise web platform (React.js, MySQL) digitising employee records, org charts, overtime approvals, IT ticketing and project scheduling.
- Engineered an automated ESD wristband monitoring gate (Arduino, C#, MySQL) enforcing 100% compliance before line entry and tracking wristband defect rates.
- Automated QR-code thermal label printing across new electronic lines through a direct C# and SQL printer interface, eliminating printing delays.
- Established centralised network file servers across all departments, removing the risk of data loss from single-drive failure.
- Recruited, mentored and scaled the software and IT engineering department from a solo role into a five-engineer team.
- Contained a worm malware outbreak across 20 laptops within 3 hours with no production downtime; sustained 99.9% uptime on core MySQL servers; resolved factory network outages inside 15-minute SLAs.

## EDUCATION

### State Polytechnic of Batam

JUL 2017 – JUL 2021

Bachelor of Applied Science (S.Tr.T.)  
Mechatronics, Robotics & Automation  
Engineering

**Graduated Cum Laude · GPA 3.81 / 4.00**

## CERTIFICATIONS

### Google Data Analytics Professional

COURSERA · 2022

### PLC Competency Certificate

BNSP INDONESIA · 2019

### Electrical Power Installation Technician

LPJK · 2020

### DevOps, Networks, Dart & Flutter

DICODING · 2023

## HONORS

2025

### 1st Place — Giken Global Value Engineering

Intelligent Machine Data: real-time  
analytics and line efficiency.

2023

### 1st Place — Giken Global Value Engineering

Manufacturing Traceability System:  
eliminated defect passage between  
processes.

## EXPERIENCE (CONTINUED)

### GIKEN PRECISION INDONESIA

Batam · Apr 2019 – Sep 2021

#### Software Test Engineer

Apr 2019 – Apr 2024

- Developed machine testing software for new electronic products and ran production trials before mass production.
- Translated customer requirements into test criteria and validated measurement accuracy between test equipment, PLCs and the MES database.

#### New Product Introduction Engineer

Apr 2019 – Sep 2021

- Studied new electronics projects from customer drawings to line readiness.
- Created and analysed FMEA, 5M change management, BOM lists and QCPC documents; ran trials and solved production problems with quality and process engineering.

### CLADTEK BI-METAL MANUFACTURING

Batam · Apr 2018 – Jan 2019

#### Junior Programmer — R&D Department

Apr 2018 – Jan 2019

- Built an autonomous pipe-inspection robot (C#, Raspberry Pi, Lidar, light sensors) navigating inside industrial pipes, streaming live video and calculating weld defect location and severity.
- Developed low-latency transmission protocols connecting embedded sensors to central servers for real-time welding QA.

## SELECTED PROJECTS

### Manufacturing Traceability System — Giken One MES

2019 – Present

Plant-wide MES with per-line production dashboards, full unit traceback and SMT refill control across 500+ product models. 1st place globally, Giken Value Engineering Award 2023.

### Intelligent Machine Data

2025

Real-time machine data acquisition and analytics replacing manual reporting with live line-efficiency and downtime visibility. 1st place globally, Giken VE Award 2025.

### Auto Feeder Machine Data Generator

2026

Digitalised the SMT feeder data flow across four roles: 5–7 hours with 4–6 people became a 30-minute automated flow, cutting cycle time by roughly 93%.

### PCBA Management Apps

2026

React.js and MySQL platform running the PCBA IT department: tickets, tasks, schedules, projects, attendance, assets and knowledge base on one live dashboard.

### SMT Pick-Up Material Analytics Dashboard

2026

Automated analysis of pick-up data across 23 SMT lines: 7.9 million pickups, error rate and spoilage PPM per line and machine, plus high-risk feeder detection.

### Digital Feeder Inventory & Tracking

2026

Real-time feeder location and asset control linking the pick-up material database to the refill system, down to area, line, machine, table and feeder number.

### LabVIEW AOI PCB Inspection System

2021

Automated Optical Inspection in LabVIEW Vision detecting missing or misaligned components on PCB assembly lines, reducing manual QA headcount.